

Project Summary

This project is a **RAG (Retrieval-Augmented Generation) based AI Teaching Assistant** designed to help students learn from their own recorded lectures or educational videos. Instead of searching through hours of video content manually, users can ask questions in natural language, such as "**In which lecture was this topic explained?**" or "**Where is binary search discussed?**" The system retrieves the most relevant portion of the uploaded lectures and generates an accurate answer based only on the user's personalized learning material.

System Workflow

The AI Teaching Assistant follows the Retrieval-Augmented Generation (RAG) pipeline:

1. **Video Upload**

Users upload one or more lecture videos to the system.

2. **Speech-to-Text Conversion**

Each video is transcribed into text using **OpenAI Whisper**, producing an accurate transcript of the lecture.

3. **Text Chunking**

The transcript is divided into smaller, meaningful chunks. This improves retrieval accuracy and allows the system to identify the most relevant sections of the lecture.

4. **Vector Embedding Generation**

Each text chunk is converted into a high-dimensional vector using the **OpenAI Embedding Model**. These embeddings capture the semantic meaning of the text and are stored in a vector database.

5. **User Query Processing**

When a user asks a question, the query is also converted into a vector using the same embedding model.

6. **Retrieval (RAG Pipeline)**

The system compares the query vector with the stored vectors and retrieves the most relevant transcript chunks using semantic similarity search.

7. **Response Generation**

The retrieved context is provided to a Large Language Model (LLM), which generates a context-aware answer. Along with the answer, the system can also indicate the lecture and timestamp where the relevant topic is discussed.

Technologies Used

- **OpenAI Whisper** – Converts lecture videos into text transcripts.
- **OpenAI Embedding Model** – Generates vector embeddings for transcript chunks and user queries.
- **Vector Database** – Stores embeddings and performs semantic similarity search.

- **Retrieval-Augmented Generation (RAG)** – Retrieves relevant context before generating responses.
- **Large Language Model (LLM)** – Produces accurate, context-aware answers based on the retrieved lecture content.

Key Features

- Upload multiple lecture or educational videos.
- Automatically generate transcripts from video content.
- Perform semantic search across all uploaded lectures.
- Answer questions using only the user's uploaded educational material.
- Identify the relevant lecture and timestamp where a topic is explained.
- Provide personalized learning assistance without requiring users to manually search through lengthy videos.